

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Eighth Semester B.E. (ISE) (Credit System) Degree Examinations

May - June 2023

19ISE41 - SOFTWARE TESTING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define software testing. With a neat diagram, explain levels of testing.	06	L*1	1	1
	b) Demonstrate with an example weak robust and strong robust testing technique.	07	L3	1	1
	c) Outline the SATM system with a neat diagram.	07	L2	1	1
2.	a) Write a short note on the followings.				
	i) Fault ii) Error iii) Test Cases iv) Incident v) Failure vi) Test	06	L1	1	1
	b) Apply the decision table technique for a commission problem.	07	L3	2	2
	c) Explain testing and debugging cycle with a neat diagram.	07	L2	2	1
3.	a) Explain the difference between black box and white box testing	06	L1	2	2
	b) Apply the decision table technique for a triangle problem.	07	L3	2	2
	c) Discuss Fault Taxonomy and give two example.	07	L2	1	1
Unit – II					
4.	a) Apply the basis path testing method for a triangle program and write the test cases.	10	L3	3	2
	b) Summarize the following terms:				
	i) Statement testing ii) Path Testing iii) Condition Testing	06	L2	3	2
	c) Interpret the test oracle with suitable diagram.	04	L2	3	2
5.	a) Apply the basis path testing method for binary search algorithm and write the test cases.	10	L3	3	2
	b) What is scaffolding? Explain the generic versus specific scaffolding.	06	L2	3	2
	c) Explain the risk management in the quality plan.	04	L2	4	2
6.	a) Apply the basis path testing method for absolute letter grading procedure, making suitable assumptions and write the test cases.	10	L3	4	2
	b) Summarize the following terms:				
	i) Risk Planning ii) Monitoring the process	06	L2	3	2
	c) Summarize chipmunk document template document title approvals.	04	L2	4	2
Unit – III					
7.	a) Outline the top-down and bottom-up approach decomposition based integration with example.	10	L2	5	1
	b) With a neat diagram explain the rapid prototyping life cycle.	10	L2	5	1
8.	a) Demonstrate the Upper level SATM finite state with neat diagram.	10	L2	4	2
	b) Interpret the following:				
	i) Acceptance Testing ii) Functional Testing iii) Integration testing				
	iv) Performance testing	10	L2	4	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
